RESULTS

This chapter presents the analysis and results of the qualitative data collected in the study in order to examine the determinants of utilization of antenatal care services among newly delivered mothers in selected hospitals in Ondo State, Nigeria. It also presents the interpretation of the results and discussion of findings. The chapter is organized into three parts. The first part presents the univariate analysis results, reporting the percent and frequency distribution of variables. The second part presents tha bivariate analysis results, reporting chi square statistic and its p-value set to statistical significance at 0.05. The third part of this section presents the multivariate analysis results, reporting odd ratio and its p-value set to statistical significance at 0.05.

The results were presented under the following headings:

4.1 Baseline and socio-demographic characteristics of participants

4.2 Maternal baseline characteristics of newly delivered mothers in selected hospitals

4.3 Assess the knowledge level of newly delivered mothers about ANC services in selected hospitals in Ondo State, Nigeria.

4.4 Determine the time of initiation of ANC among newly delivered mothers in selected hospitals in Ondo State, Nigeria

4.5 Evaluate number of ANC visits achieved by newly delivered mothers in selected hospitals in Ondo State, Nigeria

4.6 Examine the antenatal and delivery care services that are accessible to newly delivered mothers in selected hospitals in Ondo State, Nigeria

4.7 Assess the determinants of antenatal care services utilization among newly delivered mothers in selected hospitals in Ondo State, Nigeria

4.1 Baseline and socio-demographic characteristics of participants

Table 4.1: Percent and frequency distribution of study location among newly delivered mothers in selected hospitals in Ondo State, Nigeria (n = 570)

	Frequency n	Percent %
Senatorial district:		
Ondo North	167	29.3
Ondo Central	223	39.1
Ondo South	180	31.6
Local Government Area		
Akoko North East	77	13.5
Owo	90	15.8
Akure South	223	39.1
Odigbo	60	10.5
Okitipupa	120	21.1
Study Hospital name:		
State Specialist Hospital, Ikare	77	13.5
General Hospital, Owo	90	15.8
Comprehensive Health Centre, Arakale	108	18.9
Mother and Child Hospital, Akure	115	20.2
General Hospital, Ore	60	10.5
State Specialist Hospital, Okitipupa	120	21.1
	570	**100.0**

Table 4.1 presents the information on the location sites where this study was undertaken. It followed the methodology for this study and showed that the majority of newly delivered mothers who voluntarily participated in the study from selected hospitals in Ondo State were recruited from the selected hospitals in Ondo Central Senatorial District. In the same vein. 29.3% of the mothers were recruited from selected hospitals in Ondo North Senatorial District and 31.6% were recruited from selected hospitals in Ondo South Senatorial District.

Results also showed that not less than 13% of the newly delivered mothers were recruited in Akoko North East Local Government Area, even less (10.5%) were recruited in Odigbo Local

Government Area. Meanwhile, the majority (39.1%) of the newly delivered mothers were recruited in Akure South Local Government Area.

Moreover, results showed that 13.5% of the newly delivered mothers who chose to participate in this study had been recruited at State Specialist Hospital, Ikare, about 16% were recruited at General Hospital, Owo, not more than 19% were recruited at Comprehensive Health Centre, Arakale. Furthermore, a score (20.2%) were recruited at Mother and Child Hospital, Akure, meanwhile, less than that (10.5%) had been recruited at General Hospital, Ore, and 21.1% were recruited at State Specialist Hospital, Okitipupa.

Table 4.2: Socio-demographic distribution of newly delivered mothers in selected hospitals in Ondo State, Nigeria (n = 570)

	Frequency n	Percent %
Age in years: [31.8±8.9 years]		
15 – 19 years	62	10.9
20 – 24 years	91	16.0
25 – 29 years	131	23.0
30 – 34 years	71	12.5
35 – 39 years	77	13.5
40 – 44 years	62	10.9
45 – 49 years	76	13.3
Woman's highest level of education completed:		
No formal education	13	2.3
Primary	49	8.6
Secondary	289	50.7
Tertiary	219	38.4
Woman's occupational status:		
Unemployed/Unskilled	125	21.9
Semi-skilled	299	52.5
Skilled professional	146	25.6
Religion:		
Christianity: Catholic	261	45.8
Christianity: Others	241	42.3
Islam	68	11.9
Ethnic group:		
Yoruba	426	74.7
Igbo	100	17.5
Hausa	35	6.1
Other tribes	9	1.6
Marital status:		
Single	44	7.7
Married/Cohabitating	482	84.6
Separated	43	7.5
Widowed	1	0.2
Living arrangement:		
Living with partner	443	77.7
Living with own parents	58	10.2
Living with partner's parents	17	3.0
Living with other relatives	52	9.1
Partner's highest level of education completed:		
No formal education	4	0.7
Primary	12	2.1
Secondary	281	49.3
Tertiary	273	47.9
Partner's occupational status:		
Unemployed/Unskilled	82	14.4
Semiskilled	278	48.8
Skilled professional	210	36.8
Marriage type:		
Monogamy	507	88.9
Polygamy	63	11.1
Size of household:		
≤5 members	314	55.1
6 – 12 members	256	44.9
Wealth index:		
Poor	241	42.3
Moderate	275	48.2
Rich	54	9.5
	570	**100.0**

Table 4.2 presents the results on the socio-demographic characteristics of newly delivered mothers in selected hospitals in Ondo State, who participated in this study. Results showed that the mean age of newly delivered mothers who participated in this study was 31.8 years with standard deviation of 8.9 years. Meanwhile, modal age group (23.0%) was found to be at 25 – 29 years. Additionally, almost 16% of them were 20 – 24 years old, about 11% of them were 15 – 19 years old, and the oldest women in this study were 45 – 49 years old (13.3%). Results showed that a few (2.3%) of the newly delivered mothers had no formal education. However, about half (50.7%) of the mothers had secondary education, while even less (38.4%) reportedly had tertiary education. It was further revealed that more than a score (21.9%) of these newly delivered mothers were reportedly unemployed/unskilled. However, more than half (52.5%) of the mothers reported their occupations to be semi-skilled, and 25.6% reported that they had skilled professional jobs. About 9 in every 10 mothers who participated in this study were Christians – where 45.8% reported to be Catholics, and 42.3% belonged to other Christian denominations. Meanwhile only 11.9% of the mothers were reportedly Muslims.

Furthermore, results showed that the majority (74.7%) of the mothers were of the Yoruba tribe, 17.5% were of the Igbo tribe and 6.1% were of the Hausa tribe; whereas, only a few (1.6%) of the mothers reported to be from one of the minority tribes in Nigeria. Results showed that 7.7% of the newly delivered mothers were currently unmarried, 84.6% reported to be married/cohabiting with partner, only a few (0.2%) reported to be recently widowed.

Results showed that the majority (77.7%) of the newly delivered mothers who participated in this study were reportedly living with their partners. Meanwhile, not less than 10% reported to be living with their own parents. Additionally, 3.0% of them reported that they were living with partner's parents, and 9.1% of them reported that they were living with other relatives. Results

showed that a few (0.7%) of the newly delivered mothers reported that their partners had no formal education. However, just below half (49.3%) of the mothers reported that their partners had secondary education, while a little less than that (47.9%) reported that their partners had tertiary education.

It was further revealed that more than a dozen (14.4%) of these newly delivered mothers reported that their partners were unemployed/unskilled. However, less than half (48.8%) of the mothers reported their partner's occupations to be semi-skilled, and 36.8% reported that their partners had skilled professional jobs. About 9 in every 10 mothers who participated in this study were reportedly in a monogamous marriage type, meanwhile only 11.1% of the mothers were reportedly in polygamous marriage type. Furthermore, results showed that more than half (55.1%) of the mothers reported to have up to 5 members in the household, whereas, 44.9% of the mothers reported to have between 6 – 12 members in the household. Results showed that 42.3% of the newly delivered mothers reported household wealth index to be poor, 48.2% reported household wealth index to be moderate, and only some (9.5%) of the newly delivered mothers who participated in this study reported household wealth index to be rich.

4.2 Maternal baseline characteristics of newly delivered mothers in selected hospitals

Table 4.3: Baseline characteristics of newly delivered mothers in selected hospitals in Ondo State, Nigeria (n = 570)

	Frequency n	Percent %
Ever given birth:		
Yes	570	100.0
No	0	0.0
Gravidity:		
Primigravida	164	28.8
Multigravida (2-13)	406	71.2
Parity:		
Primiparous	237	41.6
Multiparous (2-12)	333	58.4
Preceding birth order:		
1st child	73	12.8
2nd child	65	11.4
3rd child	70	12.3
4th child	60	10.5
5th to 12th child	138	24.2
NA: Only child	164	28.8
Preceding birth interval (in months):		
1 – 12 months	3	0.5
13 – 24 months	61	10.7
25 – 36 months	164	28.8
37 – 48 months	79	13.9
49+ months	99	17.4
NA: Only child	164	28.8
Desire for last pregnancy:		
Wanted	267	46.8
Unwanted	303	53.2
Location of delivery:		
Respondent's home	49	8.6
Others: Home	10	1.8
Government hospital	156	27.4
Government health centre	272	47.7
Government health post	2	0.4
Others: Public sector	4	0.7
Private hospital/clinic	64	11.2
Other facilities	13	2.3
History of abortion:		
Yes	36	6.3
No	534	93.7
History of miscarriage:		
Yes	209	36.7
No	361	63.3
Mode of delivery of last birth:		
Normal vaginal delivery	558	97.9
Caesarean section	12	2.1
Mode of delivery (newborn):		
Normal vaginal delivery	533	93.5
Caesarean section	37	6.5
	570	**100.0**

Table 4.3 presents the results on baseline characteristics of newly delivered mothers in selected hospitals in Ondo State, who participated in this study. Results showed that all the newly delivered mothers who participated in this study had ever given birth, at least once. Meanwhile, not more than 29% reported to be a primigravida and about 42% were also primiparous. Additionally, 12.8% of them reported that the preceding birth was their first child, 11.4% reported the preceding birth was their second child, 12.3% reported the preceding birth was their third, 10.5% reported the preceding birth was their fourth, and 24.2% of them reported that the preceding birth was their 5th to 12th child. Results showed a few (0.5%) of the mothers reported that the least preceding birth interval they had was up to 12 months, 10.7% reported preceding birth interval to be 13 – 24 months, and 17.4% of the mothers had reported preceding birth interval to be more than 49 months. Results showed that 46.8% of the newly delivered mothers reported that they desired the last pregnancy they had, while 53.2% reported last pregnancy unwanted.

It was further revealed that 8.6% of these newly delivered mothers reported that the location of last delivery was at home or other homes (1.8%). However, 27.4% of the mothers reported that they had delivered at a Government hospital, majority (47.7%) had reportedly delivered at a Government health centre, and 0.4% reported that they delivered at a government health post or other public sector health facilities (0.7%). About 11.2% of the mothers who participated in this study reportedly delivered in a private hospital/clinic. Furthermore, results showed that very few (6.3%) of the mothers reported have no history of abortion, whereas, not more than 37% of the mothers reported had history of miscarriage. Results showed that the majority (97.9%) of the

newly delivered mothers reported they had normal vaginal delivery at last birth, and 93.5% had normal vaginal delivery for newly delivered newborn in this study.

4.3 Assess the knowledge level of newly delivered mothers about ANC services in selected hospitals in Ondo State, Nigeria

Table 4.4: Knowledge about antenatal care services among newly delivered mothers in selected hospitals in Ondo State, Nigeria (n = 570)

	Yes n (%)	No (%)	Don't know n (%)
Pregnant women should start ANC before 3months of the pregnancy	520 (91.2)	50 (8.8)	0 (0.0)
Pregnant women should attend at least 4 ANC visits	388 (68.1)	182 (31.9)	0 (0.0)
Antenatal care provides information to be able to identify danger signs during pregnancy and act promptly	508 (89.1)	62 (10.9)	0 (0.0)
Antenatal care provides the opportunity to detect early and manage pregnancy complications	515 (90.4)	55 (9.6)	0 (0.0)
It is not necessary to go for antenatal care if there is no complication during pregnancy	66 (11.6)	504 (88.4)	0 (0.0)
A woman during ANC should receive at least 2 doses of Tetanus Toxoid vaccine	5 (0.9)	565 (99.1)	0 (0.0)
Feet swelling are one of the important things that should be reported to health workers during ANC visits	7 (1.2)	563 (98.8)	0 (0.0)
Tetanus injection is given to protect mothers only from tetanus infection	42 (7.4)	528 (92.6)	0 (0.0)

Table 4.4 presents the results on knowledge about antenatal care services among the newly delivered mothers in selected hospitals in Ondo State, who participated in this study. Results showed that 91.2% of the newly delivered mothers who participated in this study had had knowledge that pregnant women should start ANC before 3 months of the pregnancy. Meanwhile, not more than 68% of the mothers reported to know that pregnant women should attend at least 4 ANC visits. Additionally, 89.1% of them reported to know that antenatal care

provides information to be able to identify danger signs during pregnancy and act promptly. The majority (90.4%) of the mothers reported knowledge that antenatal care provides information to be able to identify danger signs during pregnancy and act promptly. Results showed that 88.4 of the mothers reported that it is necessary to go for antenatal care if there is no complication during pregnancy. Meanwhile, only 0.9% reported that a woman during ANC should receive at least 2 doses of Tetanus Toxoid vaccine. It was further revealed that 98.8% of these newly delivered mothers reported that Feet swelling are NOT one of the important things that should be reported to health workers during ANC visits. However, 7.4% of the mothers reported that tetanus injection is given to protect mothers only from tetanus infection.

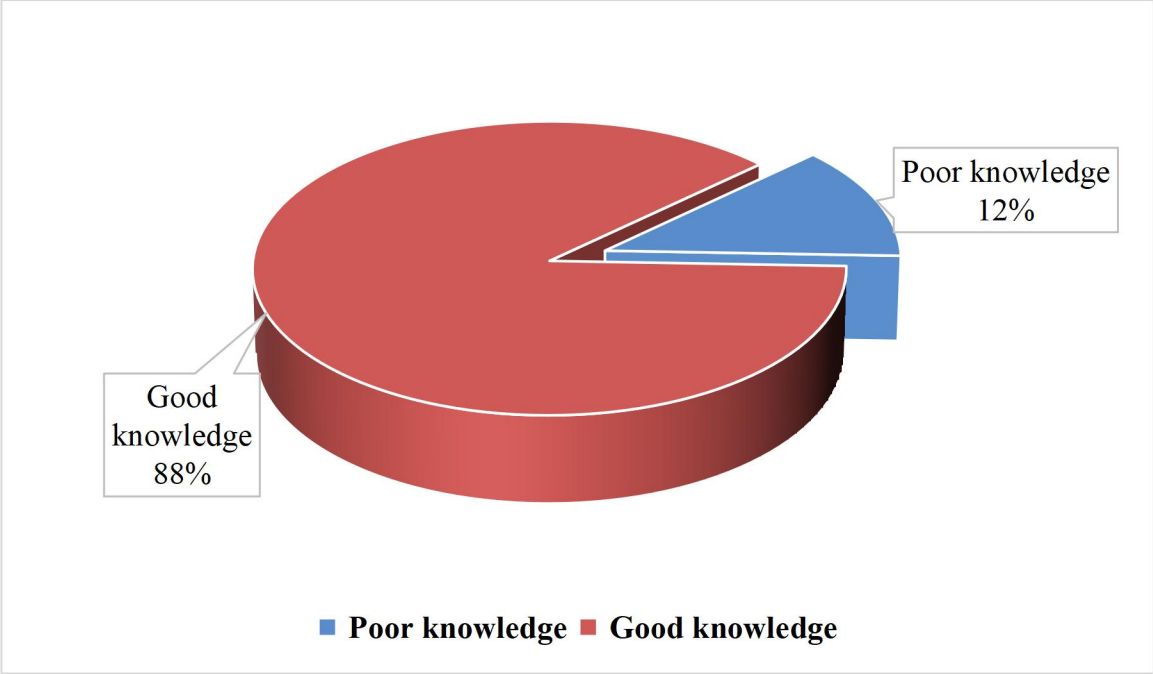

Figure 4.1: Knowledge level of newly delivered mothers about ANC services in selected hospitals in Ondo State, Nigeria

Results showed that about 88% of the newly delivered mothers had good knowledge of ANC services and 12% had poor knowledge of ANC services in selected hospitals in Ondo State, Nigeria.

4.4 Determine the time of initiation of ANC and number of ANC visits achieved by newly delivered mothers in selected hospitals in Ondo State, Nigeria

Table 4.5: The utilization of antenatal care services among newly delivered mothers in selected hospitals in Ondo State, Nigeria (n = 570)

	Frequency n	Percent %
Mother booked for antenatal:		
Yes	499	87.5
No	71	12.5
Time of initiation:		
<12 weeks of pregnancy	159	27.9
≥12 weeks of pregnancy	340	59.6
Never booked for ANC	71	12.5
Number of antenatal visits respondents attended during last pregnancy:		
<8 ANC visits	499	87.5
≥8 ANC visits	71	12.5
Pregnancy at first ANC visit during last pregnancy:		
1 – 3 months	102	17.9
4 – 6 months	354	62.1
7 – 9 months	37	6.5
Don't know	6	1.1
Never booked for ANC	71	12.5
	570	**100.0**

Table 4.5 presents the results on time of initiation of antenatal care among the newly delivered mothers in selected hospitals in Ondo State, who participated in this study. Results showed that the majority (87.5%) of the newly delivered mothers who participated in this study had registered for antenatal care service. Meanwhile, not more than 30% of the mothers reported that they booked for antenatal care when they were less than 12 weeks pregnant. Additionally, about 60% of them reported that they booked for antenatal care when they were at least, 12 weeks pregnant.

Results showed that the majority (87.5%) of the newly delivered mothers who participated in this study had attended less than 8 antenatal care visits, whereas, 12.5% of the mothers had attended

at least, 8 antenatal care visits. Meanwhile, about 18% of the mothers reported that they were 1 – 3 months pregnant at first ANC visit during last pregnancy. Additionally, 62.1% of them reported that they were 4 – 6 months pregnant at first ANC visit during last pregnancy, and few (6.5%) of the mothers reported that they were 7 – 9 months pregnant at first ANC visit during last pregnancy.

4.5 Examine the antenatal care services that are accessible to newly delivered mothers in selected hospitals in Ondo State, Nigeria

Table 4.6: Antenatal care services accessed by newly delivered mothers in selected hospitals in Ondo State, Nigeria (n = 570)

	Frequency n	Percent %
Place of antenatal care service:		
Maternity home	178	31.2
Government hospital	144	25.3
Government health centre	145	25.4
Government health post	9	1.6
Others: Public sector	23	4.0
Never booked for ANC	71	12.5
Blood pressure was measured during last pregnancy:		
Yes	331	58.1
No	168	29.5
Never booked for ANC	71	12.5
Urine sample was taken during last pregnancy:		
Yes	429	75.3
No	70	12.3
Never booked for ANC	71	12.5
Blood sample was taken during last pregnancy:		
Yes	459	80.5
No	40	7.0
Never booked for ANC	71	12.5
Received tetanus toxoid injection during last pregnancy:		
Yes	445	78.1
No	47	8.2
Don't know	7	1.2
Never booked for ANC	71	12.5
Doses of tetanus toxoid injection received during last pregnancy:		
Received no injection	46	8.1
1 dose	119	20.9
2 doses	327	57.4
Don't know	7	1.2
Never booked for ANC	71	12.5
Mother was given/bought any of iron and folic acid tablets or iron syrup during last pregnancy:		
Yes	456	80.0
No	41	7.2
Don't know	2	0.4
Never booked for ANC	71	12.5
Number of iron and folic acid tablets prescribed:		
<10 packs	1	0.2
≥10 packs	430	75.4
Don't know	68	11.9
Never booked for ANC	71	12.5
	570	**100.0**

Table 4.6 presents the results on the antenatal care services accessed by the newly delivered mothers in selected hospitals in Ondo State, who participated in this study. Results showed that the majority (31.2%) of the newly delivered mothers who participated in this study had received antenatal care services at maternity homes, whereas 25.3% of the mothers had received antenatal care services at a government hospital, 25.4% had received antenatal care services at a government health centre. Meanwhile, 58.1% of the mothers reported that their blood pressure was measured at the antenatal visits during last pregnancy. Additionally, 75.3% of the mothers reported that their urine sample was taken at the antenatal visits during last pregnancy, and 80.5% of the mothers reported that their blood sample was taken at the antenatal visits during last pregnancy.

Moreover, most (78.1%) of the mothers reported that they received tetanus toxoid injection at the antenatal visits during last pregnancy, and more than half (57.4%) had reportedly received 2 doses of tetanus toxoid injection. Results also showed that the majority (80.0%) of the mothers had been given/bought iron and folic acid tablets or iron syrup during last pregnancy, while around three-quarters of them took at least 10 packs as prescribed.

4.6 Assess the determinants of antenatal care services utilization among newly delivered mothers in selected hospitals in Ondo State, Nigeria

Table 4.7: Study location as a determinant of antenatal care services utilization among newly delivered mothers in selected hospitals in Ondo State, Nigeria (n = 570)

		Antenatal care services utilized						
	ANC utilization %	ANC initiation at <12 weeks %	<8 ANC visits %	Blood sample taken %	Received Tetanus injection %	Received 2 doses of tetanus injection %	Received Iron & Folic acid tabs/syrup %	Number of women n
Senatorial district:	*0.302*	*0.146*	*0.778*	*0.208*	*0.445*	*0.220*	*0.657*	
Ondo North	90.4	34.7	63.5	80.2	81.4	57.5	81.4	167
Ondo Central	85.2	24.2	57.4	78.9	77.1	61.0	78.5	223
Ondo South	87.8	26.1	59.4	82.8	76.1	52.8	80.6	180
Local Government Area	*0.349*	*0.161*	*0.908*	*0.417*	*0.398*	*0.253*	*0.659*	
Akoko North East	92.2	39.0	64.9	81.8	83.1	62.3	83.1	77
Owo	88.9	31.1	62.2	78.9	80.0	53.3	80.0	90
Akure South	85.2	24.2	57.4	78.9	77.1	61.0	78.5	223
Odigbo	83.3	18.3	56.7	80.0	68.3	48.3	75.0	60
Okitipupa	90.0	30.0	60.8	84.2	80.0	55.0	83.3	120
Study Hospital name:	*0.402*	*0.243*	*0.951*	*0.479*	*0.486*	*0.360*	*0.703*	
State Specialist Hospital, Ikare	92.2	39.0	64.9	81.8	83.1	62.3	83.1	77
General Hospital, Owo	88.9	31.1	62.2	78.9	80.0	53.3	80.0	90
Comprehensive Health Centre, Arakale	83.3	25.0	55.6	78.7	75.9	58.3	76.9	108
Mother and Child Hospital, Akure	87.0	23.5	59.1	79.1	78.3	63.5	80.0	115
General Hospital, Ore	83.3	18.3	46.7	80.0	68.3	48.3	75.0	60
State Specialist Hospital, Okitipupa	90.0	30.0	60.8	84.2	80.0	55.0	83.3	120
Total	**87.5**	**27.9**	**59.8**	**80.5**	**78.1**	**57.4**	**80.0**	**570**

Table 4.7 presents the results on the study location determinants of antenatal care services utilization among the newly delivered mothers in selected hospitals in Ondo State, who participated in this study. Results showed that the uptake of antenatal care services utilization was 90.4% in Ondo North Senatorial district, 85.2% in Ondo Central Senatorial district, and 87.5% in Ondo South Senatorial district. However, there was no significant association between Senatorial district zoning and uptake of antenatal care services utilization during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.302$). Results further showed that the uptake of antenatal care services utilization was 92.2% in Akoko North Local Government Area, 88.9% in Owo Local Government Area, 85.2% in Akure South Local Government Area, 83.3% in Odigbo Local Government Area, and 90.0% in Okitipupa Local Government Area. Similarly, there was no significant association between Local Government Area and uptake of antenatal care services utilization during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.349$). Results also revealed that the uptake of antenatal care services utilization was 92.2% in State Specialist Hospital, Ikare, 88.9% in General Hospital, Owo, 83.3% in Comprehensive Health Centre, Arakale, and General Hospital, Ore, 87.0% in Mother and Child Hospital, Akure, and 90.0% in State Specialist Hospital, Okitipupa. Similarly, there was significant association between study hospital site and uptake of antenatal care services utilization during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.402$).

Results further showed that the pattern of ANC initiation below 12 weeks of pregnancy was 34.7% in Ondo North Senatorial district, 24.2% in Ondo Central Senatorial district, and 26.1% in Ondo South Senatorial district. However, there was no significant association between Senatorial district zoning and pattern of ANC initiation below 12 weeks of pregnancy during antenatal care

visits among newly delivered mothers who participated in this study (p = 0.146). Results further showed that the pattern of ANC initiation below 12 weeks of pregnancy was 39.0% in Akoko North Local Government Area, 31.1% in Owo Local Government Area, 24.2% in Akure South Local Government Area, 18.3% in Odigbo Local Government Area, and 30.0% in Okitipupa Local Government Area. Similarly, there was no significant association between Local Government Area and pattern of ANC initiation below 12 weeks of pregnancy during antenatal care visits among newly delivered mothers who participated in this study (p = 0.161). Results also showed that the pattern of ANC initiation below 12 weeks of pregnancy was 39.0% in State Specialist Hospital, Ikare, 31.1% in General Hospital, Owo, 25.0% in Comprehensive Health Centre, Arakale, 23.5% in Mother and Child Hospital, Akure, 18.3% in and General Hospital, Ore, and 30.0% in State Specialist Hospital, Okitipupa. Similarly, there was no significant association between study hospital site and pattern of ANC initiation below 12 weeks of pregnancy during antenatal care visits among newly delivered mothers who participated in this study (p = 0.243).

Results further showed that the pattern of ANC visits less than 8 was 63.5% in Ondo North Senatorial district, 57.4% in Ondo Central Senatorial district, and 59.4% in Ondo South Senatorial district. However, there was no significant association between Senatorial district zoning and pattern of ANC visits less than 8 during antenatal care visits among newly delivered mothers who participated in this study (p = 0.778). Results further showed that the pattern of ANC visits less than 8 was 64.9% in Akoko North Local Government Area, 62.2% in Owo Local Government Area, 57.4% in Akure South Local Government Area, 56.7% in Odigbo Local Government Area, and 60.8% in Okitipupa Local Government Area. Similarly, there was no significant association between Local Government Area and pattern of ANC visits less than 8 during antenatal care visits among newly delivered mothers who participated in this study (p =

0.908). Results also showed that the pattern of ANC visits less than 8 was 64.9% in State Specialist Hospital, Ikare, 62.2% in General Hospital, Owo, 55.6% in Comprehensive Health Centre, Arakale, 59.1% in Mother and Child Hospital, Akure, 46.7% in and General Hospital, Ore, and 60.8% in State Specialist Hospital, Okitipupa. Similarly, there was no significant association between study hospital site and pattern of ANC visits less than 8 during antenatal care visits among newly delivered mothers who participated in this study (p = 0.951).

Results further showed that the pattern of blood sample taken was 80.2% in Ondo North Senatorial district, 78.9% in Ondo Central Senatorial district, and 82.8% in Ondo South Senatorial district. However, there was no significant association between Senatorial district zoning and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.208). Results further showed that the pattern of blood sample taken was 81.8% in Akoko North Local Government Area, 78.9% in Owo Local Government Area, 78.9% in Akure South Local Government Area, 80.0% in Odigbo Local Government Area, and 84.2% in Okitipupa Local Government Area. There was significant no association between Local Government Area and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.417). Results also showed that the pattern of blood sample taken was 81.8% in State Specialist Hospital, Ikare, 78.9% in General Hospital, Owo, 78.7% in Comprehensive Health Centre, Arakale, 79.1% in Mother and Child Hospital, Akure, 80.0% in and General Hospital, Ore, and 84.2% in State Specialist Hospital, Okitipupa. Similarly, there was no significant association between study hospital site and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.479).

Results further showed that the pattern of tetanus injection received was 81.4% in Ondo North Senatorial district, 77.1% in Ondo Central Senatorial district, and 76.1% in Ondo South Senatorial district. However, there was no significant association between Senatorial district zoning and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.445). Results further showed that the pattern of tetanus injection received was 83.1% in Akoko North Local Government Area, 80.0% in Owo Local Government Area, 77.1% in Akure South Local Government Area, 68.3% in Odigbo Local Government Area, and 80.0% in Okitipupa Local Government Area. There was significant no association between Local Government Area and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.398). Results also showed that the pattern of tetanus injection received was 83.1% in State Specialist Hospital, Ikare, 80.0% in General Hospital, Owo, 75.9% in Comprehensive Health Centre, Arakale, 78.3% in Mother and Child Hospital, Akure, 68.3% in and General Hospital, Ore, and 80.0% in State Specialist Hospital, Okitipupa. Similarly, there was no significant association between study hospital site and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.486).

Moreover, results showed that the pattern of Iron & Folic acid tabs/syrup received was 81.4% in Ondo North Senatorial district, 78.5% in Ondo Central Senatorial district, and 80.6% in Ondo South Senatorial district. However, there was no significant association between Senatorial district zoning and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.657). Results further showed that the pattern of Iron & Folic acid tabs/syrup received was 83.1% in Akoko North Local Government Area, 80.0% in Owo Local Government Area, 78.5% in Akure South Local

Government Area, 75.0% in Odigbo Local Government Area, and 83.3 in Okitipupa Local Government Area. There was significant no association between Local Government Area and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.659$). Results also showed that the pattern of Iron & Folic acid tabs/syrup received was 83.1% in State Specialist Hospital, Ikare, 80.0% in General Hospital, Owo, 76.9% in Comprehensive Health Centre, Arakale, 80.0% in Mother and Child Hospital, Akure, 75.0% in and General Hospital, Ore, and 83.3% in State Specialist Hospital, Okitipupa. Similarly, there was no significant association between study hospital site and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.703$).

Table 4.8: Socio-demographic distribution as a determinant of antenatal care services utilization among newly delivered mothers in selected hospitals in Ondo State, Nigeria (n = 570)

		Antenatal care services utilized						
	ANC utilization %	Bp measured %	Urine sample taken %	Blood sample taken %	Received Tetanus injection %	Received 2 doses of tetanus injection %	Received Iron & Folic acid tabs/syrup %	Number of women n
Age in years:	*0.388*	*0.077*	*0.408*	*0.514*	*0.532*	*0.773*	*0.203*	
15 – 19 years	83.5	49.5	68.0	75.7	74.8	51.5	78.6	62
20 – 24 years	85.9	60.0	72.9	76.5	82.4	61.2	76.5	91
25 – 29 years	84.4	68.8	76.0	78.1	76.0	57.3	82.3	131
30 – 34 years	93.0	59.2	80.3	83.1	78.9	53.5	80.3	71
35 – 39 years	89.6	61.0	81.8	87.0	83.1	63.6	84.4	77
40 – 44 years	87.1	53.2	69.4	79.0	72.6	58.1	74.2	62
45 – 49 years	92.1	53.9	80.3	86.8	78.9	57.9	82.9	76
Woman's highest level of education completed:	*0.461*	*0.299*	*0.179*	*0.654*	*0.945*	*0.513*	*0.721*	
No formal education	100.0	84.6	69.2	100.0	92.3	46.2	100.0	13
Primary	87.8	55.1	69.4	79.6	77.6	67.3	75.5	49
Secondary	86.2	59.5	76.1	79.6	76.8	56.7	79.2	289
Tertiary	88.6	55.3	75.8	80.8	79.0	56.6	80.8	219
Woman's occupational status:	*0.778*	*0.562*	*0.322*	*0.916*	*0.661*	*0.795*	*0.766*	
Unemployed/Unskilled	88.8	55.2	72.0	81.6	78.4	58.4	78.4	125
Semiskilled	86.6	60.2	76.9	80.3	77.6	56.2	80.3	299
Skilled professional	88.4	56.2	74.7	80.1	78.8	58.9	80.8	146
Religion:	*0.067*	*0.244*	*0.081*	*0.073*	*0.282*	*0.414*	*0.438*	
Christianity: Catholic	87.7	58.6	77.8	79.3	77.8	57.9	80.5	261
Christianity: Others	85.1	56.0	70.5	78.4	76.3	54.8	77.6	241
Islam	95.6	63.2	82.4	92.6	85.3	64.7	86.8	68
Ethnic group:	*0.200*	*0.214*	*0.233*	*0.162*	***0.009***	***0.024***	*0.553*	
Yoruba	86.9	56.6	73.7	79.3	78.9	58.7	78.9	426
Igbo	86.0	58.0	76.0	78.0	74.0	53.0	79.0	100
Hausa	97.1	77.1	91.4	97.1	85.7	57.1	91.4	35
Other tribes	100.0	55.6	77.8	100.0	55.6	44.4	100.0	9

Marital status:	*0.884*	*0.470*	*0.883*	*0.974*	*0.974*	*0.993*	*0.878*	
Single	88.6	52.3	77.3	84.1	77.3	56.8	79.5	44
Married/Cohabitating	87.1	59.3	75.3	79.9	77.8	57.1	80.3	482
Separated	90.7	48.8	72.1	83.7	84.4	60.5	76.7	43
Widowed	100.0	100.0	100.0	100.0	100.0	100.0	100.0	1
Total	**87.5**	**58.1**	**75.3**	**80.5**	**78.1**	**57.4**	**80.0**	**570**

Table 4.8 presents the results on the sociodemographic determinants of antenatal care services utilization among the newly delivered mothers in selected hospitals in Ondo State, who participated in this study. Results showed that the uptake of antenatal care services utilization was 83.3% among mothers in 15 – 19 years age group, 93.0% among mothers in 30 – 34 years age group, and 92.1% among mothers in 45 – 49 years age group. There was no significant association between age and uptake of antenatal care services utilization during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.388$). Results further showed that the uptake of antenatal care services utilization was reported in 100.0% of the mothers with no formal education, 87.8% among women with primary education, 86.2% among women with secondary education, and 88.6% among women with tertiary education. Similarly, there was no significant association between highest level of education completed and uptake of antenatal care services utilization during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.461$). Results also revealed that the uptake of antenatal care services utilization was 86.9% among mothers of Yoruba ethnic group, 86.0% among mothers of Igbo ethnic group, 97.1% among mothers of Hausa ethnic group, and reported in 100.0% of the mothers of other ethnic groups. Similarly, there was no significant association between ethnicity and uptake of antenatal care services utilization during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.200$). Results also showed that the uptake of antenatal care services utilization was 88.6% among single mothers, 87.1% among married/cohabiting mothers, 90.7% among separated mothers, and all 100.0% of the widowed mothers. Similarly, there was no significant association between marital status and uptake of antenatal care services utilization during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.884$).

Results further showed that the pattern of blood pressure measurement was 49.5% among mothers in 15 – 19 years age group, 59.2% among mothers in 30 – 34 years age group, and 53.9% among mothers in 45 – 49 years age group. There was no significant association between age and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study (p = 0.077). Results further showed that the pattern of blood pressure measurement was 84.6% among mothers with no formal education, 55.1% among women with primary education, 59.5% among women with secondary education, and 55.3% among women with tertiary education. Similarly, there was no significant association between highest level of education completed and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study (p = 0.299). Results also showed that the pattern of blood pressure measurement was 56.6% among mothers of Yoruba ethnic group, 58.0% among mothers of Igbo ethnic group, 77.1% among mothers of Hausa ethnic group, and 55.6% among mothers of other ethnic groups. Similarly, there was no significant association between ethnicity and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study (p = 0.214). Results also showed that the pattern of blood pressure measurement was 52.3% among single mothers, 59.3% among married/cohabiting mothers, 48.8% among separated mothers, and all 100.0% of the widowed mothers. Similarly, there was no significant association between marital status and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study (p = 0.470).

Results further showed that the pattern of urine sample taken was 68.0% among mothers in 15 – 19 years age group, 80.3% among mothers in 30 – 34 years age group, and 80.3% among mothers in 45 – 49 years age group. There was no significant association between age and

pattern of urine sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.408). Results further showed that the pattern of urine sample taken was 69.2% among mothers with no formal education, 69.4% among women with primary education, 76.1% among women with secondary education, and 75.8% among women with tertiary education. Similarly, there was no significant association between highest level of education completed and pattern of urine sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.179). Results also showed that the pattern of urine sample taken was 73.7% among mothers of Yoruba ethnic group, 76.0% among mothers of Igbo ethnic group, 91.4% among mothers of Hausa ethnic group, and 77.8% among mothers of other ethnic groups. Similarly, there was no significant association between ethnicity and pattern of urine sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.233). Results also showed that the pattern of urine sample taken was 77.3% among single mothers, 75.3% among married/cohabiting mothers, 72.1% among separated mothers, and all 100.0% of the widowed mothers. Similarly, there was no significant association between marital status and pattern of urine sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.883).

Results further showed that the pattern of blood sample taken was 75.7% among mothers in 15 – 19 years age group, 83.1% among mothers in 30 – 34 years age group, and 86.8% among mothers in 45 – 49 years age group. There was no significant association between age and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.514). Results further showed that the pattern of blood sample taken was reported in 100.0% of the mothers with no formal education, 79.6% among women with primary education, 79.6% among women with secondary education, and 80.8% among

women with tertiary education. Similarly, there was no significant association between highest level of education completed and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.654). Results also showed that the pattern of blood sample taken was 79.3% among mothers of Yoruba ethnic group, 78.0% among mothers of Igbo ethnic group, 97.1% among mothers of Hausa ethnic group, and reported in 100.0% of the mothers of other ethnic groups. Similarly, there was no significant no association between ethnicity and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.162). Results also showed that the pattern of blood sample taken was 84.1% among single mothers, 79.9% among married/cohabiting mothers, 83.7% among separated mothers, and all 100.0% of the widowed mothers. Similarly, there was no significant association between marital status and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.974).

Results further showed that the pattern of tetanus injection received was 74.8% among mothers in 15 – 19 years age group, 78.9% among mothers in 30 – 34 years age group, and 78.9% among mothers in 45 – 49 years age group. There was no significant association between age and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.532). Results further showed that the pattern of tetanus injection received was 92.3% among mothers with no formal education, 77.6% among women with primary education, 76.8% among women with secondary education, and 79.0% among women with tertiary education. Similarly, there was no significant association between highest level of education completed and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.945). Results also showed

that the pattern of tetanus injection received was 78.9% among mothers of Yoruba ethnic group, 74.0% among mothers of Igbo ethnic group, 85.7% among mothers of Hausa ethnic group, and 55.6% among mothers of other ethnic groups. Similarly, there was significant association between ethnicity and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.009). Results also showed that the pattern of tetanus injection received was 77.3% among single mothers, 77.8% among married/cohabiting mothers, 84.4% among separated mothers, and all 100.0% of the widowed mothers. Similarly, there was no significant association between marital status and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.974).

Results further showed that the pattern of Iron & Folic acid tabs/syrup received was 78.6% among mothers in 15 – 19 years age group, 80.3% among mothers in 30 – 34 years age group, and 82.9% among mothers in 45 – 49 years age group. There was no significant association between age and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.203). Results further showed that the pattern of Iron & Folic acid tabs/syrup received was reported in 100.0% of the mothers with no formal education, 75.5% among women with primary education, 79.2% among women with secondary education, and 80.8% among women with tertiary education. Similarly, there was no significant association between highest level of education completed and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.721). Results also showed that the pattern of Iron & Folic acid tabs/syrup received was 78.9% among mothers of Yoruba ethnic group, 79.0% among mothers of Igbo ethnic group, 91.4% among mothers of Hausa ethnic group, and reported in

100.0% of the mothers of other ethnic groups. Similarly, there was no significant association between ethnicity and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.553$). Results also showed that the pattern of Iron & Folic acid tabs/syrup received was 79.5% among single mothers, 80.3% among married/cohabiting mothers, 76.7% among separated mothers, and all 100.0% of the widowed mothers. Similarly, there was no significant association between marital status and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.878$).

Table 4.9: Household characteristics as a determinant of antenatal care services utilization among newly delivered mothers in selected hospitals in Ondo State, Nigeria (n = 570)

		Antenatal care services utilized						
	ANC utilization %	Bp measured %	Urine sample taken %	Blood sample taken %	Received Tetanus injection %	Received 2 doses of tetanus injection %	Received Iron & Folic acid tabs/syrup %	Number of women n
Living arrangement:	*0.438*	*0.424*	*0.405*	*0.618*	*0.840*	*0.710*	*0.828*	
Living with partner	87.1	58.2	74.5	79.5	78.1	56.4	79.2	443
Living with own parents	86.2	50.0	70.7	81.0	75.9	55.2	77.6	58
Living with partner's parents	82.4	47.1	70.6	76.5	76.5	55.9	82.4	17
Living with other relatives	94.2	69.2	88.5	90.4	80.8	69.2	88.5	52
Partner's highest level of education completed:	*0.610*	*0.801*	*0.741*	*0.667*	*0.490*	*0.601*	*0.983*	
No formal education	100.0	75.0	100.0	100.0	100.0	50.0	100.0	4
Primary	83.3	66.7	75.0	83.3	75.0	50.0	75.0	12
Secondary	86.1	55.9	75.1	80.1	76.9	56.6	79.0	281
Tertiary	89.0	59.7	75.1	80.6	79.1	58.6	81.0	273
Partner's occupational status:	*0.231*	*0.560*	*0.387*	*0.404*	*0.447*	*0.634*	*0.494*	
Unemployed/Unskilled	91.5	59.8	76.8	85.4	82.9	61.0	82.9	82
Semiskilled	85.3	56.5	74.8	79.1	75.2	54.3	78.1	278
Skilled professional	89.0	59.5	75.2	80.5	80.0	60.0	81.4	210
Marriage type:	*0.156*	*0.205*	*0.260*	***0.044***	*0.260*	*0.413*	*0.406*	
Monogamy	86.8	57.0	74.4	79.1	76.9	56.4	79.1	507
Polygamy	93.7	66.7	82.5	92.1	87.3	65.1	87.3	63
Size of household:	*0.055*	***<0.0001***	*0.131*	***0.041***	***0.022***	***0.021***	***0.020***	
≤5 members	85.0	48.4	72.9	76.8	73.2	51.6	75.5	314
6 – 12 members	90.6	69.9	78.1	85.2	84.0	64.5	85.5	256
Wealth index:	*0.755*	*0.695*	*0.713*	*0.917*	*0.744*	*0.827*	*0.942*	
Poor	87.1	59.8	73.0	80.9	77.2	58.1	78.4	241
Moderate	87.3	57.5	76.7	79.6	77.8	55.6	80.4	275
Rich	90.7	53.7	77.8	83.3	83.3	63.0	85.2	54
Total	**87.5**	**58.1**	**75.3**	**80.5**	**78.1**	**57.4**	**80.0**	**570**

Table 4.9 presents the results on the household characteristics as determinants of antenatal care services utilization among the newly delivered mothers in selected hospitals in Ondo State, who participated in this study. Results showed that the uptake of antenatal care services utilization was 87.1% among mothers living with their partners, 86.2% among mothers living with own parents, and 94.2% among mothers living with other relatives. There was no significant association between living arrangement and uptake of antenatal care services utilization during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.438$). Results further showed that the uptake of antenatal care services utilization was reported in 100.0% of the mothers with partners who had no formal education, 83.3% among women with partners who had primary education, 86.1% among women with partners who had secondary education, and 89.0% among women with partners who had tertiary education. Similarly, there was no significant association between partner's highest level of education completed and uptake of antenatal care services utilization during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.610$). Results also revealed that the uptake of antenatal care services utilization was 85.0% among mothers with household size up to 5 members, and 90.6% of the mothers with household size of more than 5 members. Similarly, there was no significant association between size of household and uptake of antenatal care services utilization during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.055$). Results also showed that the uptake of antenatal care services utilization was 87.1% among mothers in poor households, 87.3% among mothers in moderately rich households, and 90.7% of the mothers in rich households. Similarly, there was no significant association between household wealth index and uptake of antenatal care services utilization during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.755$).

Results further showed that the pattern of blood pressure measurement was 58.2% among mothers living with their partners, 50.0% among mothers living with own parents, and 69.2% among mothers living with other relatives. There was no significant association between living arrangement and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.424$). Results further showed that the pattern of blood pressure measurement was 75.0% of the mothers with partners who had no formal education, 66.7% among women with partners who had primary education, 55.9% among women with partners who had secondary education, and 59.7% among women with partners who had tertiary education. Similarly, there was no significant association between partner's highest level of education completed and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.801$). Results also showed that the pattern of blood pressure measurement was 48.4% among mothers with household size up to 5 members, and 69.9% of the mothers with household size of more than 5 members. Similarly, there was significant association between size of household and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p < 0.0001$). Results also showed that the pattern of blood pressure measurement was 59.8% among mothers in poor households, 57.5% among mothers in moderately rich households, and 53.7% of the mothers in rich households. Similarly, there was no significant association between household wealth index and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.695$).

Results further showed that the pattern of urine sample taken was 74.5% among mothers living with their partners, 70.7% among mothers living with own parents, and 88.5% among mothers

living with other relatives. There was no significant association between living arrangement and pattern of urine sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.405). Results further showed that the pattern of urine sample taken was reported in 100.0% of the mothers with partners who had no formal education, 75.0% among women with partners who had primary education, 75.1% among women with partners who had secondary education, and 75.1% among women with partners who had tertiary education. Similarly, there was no significant association between partner's highest level of education completed and pattern of urine sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.741). Results also showed that the pattern of urine sample taken was 72.9% among mothers with household size up to 5 members, and 78.1% of the mothers with household size of more than 5 members. Similarly, there was no significant association between size of household and pattern of urine sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.131). Results also showed that the pattern of urine sample taken was 73.0% among mothers in poor households, 76.7% among mothers in moderately rich households, and 77.8% of the mothers in rich households. Similarly, there was no significant association between household wealth index and pattern of urine sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.713).

Results further showed that the pattern of blood sample taken was 79.5% among mothers living with their partners, 81.0% among mothers living with own parents, and 90.4% among mothers living with other relatives. There was no significant association between living arrangement and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.618). Results further showed that the pattern of blood sample

taken was reported in 100.0% of the mothers with partners who had no formal education, 83.3% among women with partners who had primary education, 80.1% among women with partners who had secondary education, and 80.6% among women with partners who had tertiary education. Similarly, there was no significant association between partner's highest level of education completed and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.667). Results also showed that the pattern of blood sample taken was 76.8% among mothers with household size up to 5 members, and 85.2% of the mothers with household size of more than 5 members. Similarly, there was significant association between size of household and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.041). Results also showed that the pattern of blood sample taken was 80.9% among mothers in poor households, 79.6% among mothers in moderately rich households, and 83.3% of the mothers in rich households. Similarly, there was no significant association between household wealth index and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.917).

Results further showed that the pattern of tetanus injection received was 78.1% among mothers living with their partners, 75.9% among mothers living with own parents, and 80.8% among mothers living with other relatives. There was no significant association between living arrangement and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.840). Results further showed that the pattern of tetanus injection received was reported in 100.0% of the mothers with partners who had no formal education, 75.0% among women with partners who had primary education, 76.9% among women with partners who had secondary education, and 79.1% among women with

partners who had tertiary education. Similarly, there was no significant association between partner's highest level of education completed and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.490). Results also showed that the pattern of tetanus injection received was 73.2% among mothers with household size up to 5 members, and 84.0% of the mothers with household size of more than 5 members. Similarly, there was significant association between size of household and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.022). Results also showed that the pattern of tetanus injection received was 77.2% among mothers in poor households, 77.8% among mothers in moderately rich households, and 83.3% of the mothers in rich households. Similarly, there was no significant association between household wealth index and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.744).

Results further showed that the pattern of Iron & Folic acid tabs/syrup received was 79.2% among mothers living with their partners, 77.6% among mothers living with own parents, and 80.5% among mothers living with other relatives. There was no significant association between living arrangement and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.828). Results further showed that the pattern of Iron & Folic acid tabs/syrup received was reported in 100.0% of the mothers with partners who had no formal education, 75.0% among women with partners who had primary education, 79.0% among women with partners who had secondary education, and 81.4% among women with partners who had tertiary education. Similarly, there was no significant association between partner's highest level of education completed and pattern of Iron

& Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.983). Results also showed that the pattern of Iron & Folic acid tabs/syrup received was 75.5% among mothers with household size up to 5 members, and 85.5% of the mothers with household size of more than 5 members. Similarly, there was significant association between size of household and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.020). Results also showed that the pattern of Iron & Folic acid tabs/syrup received was 78.4% among mothers in poor households, 80.4% among mothers in moderately rich households, and 85.2% of the mothers in rich households. Similarly, there was no significant association between household wealth index and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.942).

Table 4.10: Binary logistic regression modelling of having good knowledge of antenatal care services and socio-demographic factors among newly delivered mothers in selected hospitals in Ondo State, Nigeria (n = 570)

	Knowledge level of ANC Services: Good		
Variables	**Odds Ratio**	**[95% C.I OR]**	**p-value**
Age in years:			
15 – 19 years	0.355	[0.59 – 3.47]	0.434
20 – 24 years	0.564	[0.65 – 4.77]	0.268
25 – 29 years	0.587	[0.67 – 4.81]	0.243
30 – 34 years	0.003	[0.39 – 2.59]	0.994
35 – 39 years	0.287	[0.51 – 3.50]	0.561
40 – 44 years	-0.420	[0.26 – 1.66]	0.375
45 – 49 years	-	-	-
Woman's highest level of education completed:			
No formal education	-0.610	[0.11 – 2.77]	0.463
Primary	0.910	[0.57 – 10.9]	0.228
Secondary	-0.126	[0.21 – 3.79]	0.866
Tertiary	-	-	-
Woman's occupational status:			
Unemployed/Unskilled	0.636	[0.73 – 4.91]	0.192
Semiskilled	0.825	[0.51 – 10.1]	0.278
Skilled professional	-	-	-
Religion:			
Christianity: Catholic	-0.532	[0.13 -2.67]	0.491
Christianity: Others	-1.563	[0.05 – 0.95]	**0.043**
Islam	-	-	-
Ethnic group:			
Yoruba	0.400	[0.28 – 8.09]	0.643
Igbo	0.834	[0.37 – 14.2]	0.370
Hausa	-0.148	[0.07 – 10.9]	0.909
Other tribes	-	-	-
Marital status:			
Single	-18.9	[0 – na]	1.000
Married/Cohabitating	-19.0	[0 – na]	1.000
Separated	-18.2	[0 – na]	1.000
Widowed	-	-	-

Table 4.10 presents the results on the binary logistic regression modelling of having good knowledge of antenatal care services and socio-demographic factors among the newly delivered mothers in selected hospitals in Ondo State, who participated in this study. The results showed that for every participant in 15 - 19 years age group, the odds of having good knowledge of ANC

services compared to oldest age groups insignificantly decreases by 65.5% (OR = 0.355, p = 0.434); for every participant in 30 - 34 years age group, the odds of having good knowledge of ANC services compared to oldest age groups insignificantly decreases by 99.7% (OR = 0.003, p = 0.994); and for every participant in 40 - 44 years age group, the odds of having good knowledge of ANC services compared to oldest age groups insignificantly decreases by 58.0% (OR = -0.420, p = 0.375).

It can also be seen that for every participant with no formal education, the odds of having good knowledge of ANC services compared to the participant with tertiary education insignificantly decreases by 39% (OR = -0.610, p = 0.463); for every participant with primary education, the odds of having good knowledge of ANC services compared to the participant with tertiary education insignificantly decreases by 9% (OR = 0.910, p = 0.228); and for every participant with secondary education, the odds of having good knowledge of ANC services compared to the participant with tertiary education insignificantly decreases by 87.4% (OR = -0.126, p = 0.866).

Results further showed that for every participant who were unemployed/unskilled, the odds of having good knowledge of ANC services compared to the participant who were professionally skilled insignificantly decreases by 36.4% (OR = 0.636, p = 0.192); and for every participant who were semi-skilled, the odds of having good knowledge of ANC services compared to the participant who were professionally skilled insignificantly decreases by 17.5% (OR = 0.825, p = 0.278).

Moreover, results showed that for every Catholic-Chirstian mothers who participate in this study, the odds of having good knowledge of ANC services compared to those who were Muslims insignificantly decreases by 46.8% (OR = -0.532, p = 0.491); and for every Chirstian mothers of other denominations who participate in this study, the odds of having good knowledge of ANC

services compared to those who were Muslims significantly increases by 56.3% (OR = -1.563, p = 0.043).

Results further showed that for every Yoruba participant, the odds of having good knowledge of ANC services compared to the participants of other tribes insignificantly decreases by 60.0% (OR = 0.400, p = 0.643); for every Igbo participant, the odds of having good knowledge of ANC services compared to the participants of other tribes insignificantly decreases by 16.6% (OR = 0.834, p = 0.370); and for every Hausa participant, the odds of having good knowledge of ANC services compared to the participants of other tribes insignificantly decreases by 85.2% (OR = -0.148, p = 0.909).

Table 4.11: Binary logistic regression modelling of having good knowledge of antenatal care services and socio-demographic factors among newly delivered mothers in selected hospitals in Ondo State, Nigeria (n = 570)

	Knowledge level of ANC Services: Good		
Variables	**Odds Ratio**	**[95% C.I OR]**	**p-value**
Living arrangement:			
Living with partner	-0.751	[0.14 – 1.60]	0.227
Living with own parents	-0.742	[0.11 – 2.00]	0.312
Living with partner's parents	-1.140	[0.06 – 1.82]	0.198
Living with other relatives	-	-	-
Partner's highest level of education completed:			
No formal education	18.2	[0 – na]	0.999
Primary	18.5	[0 – na]	0.999
Secondary	-0.286	[0.27 – 2.09]	0.584
Tertiary	-	-	-
Partner's occupational status:			
Unemployed/Unskilled	1.044	[0.78 – 10.3]	0.113
Semiskilled	0.655	[0.67 – 5.57]	0.227
Skilled professional	-	-	-
Marriage type:			
Monogamy	-0.525	[0.24 – 1.48]	0.262
Polygamy	-	-	-
Size of household:			
≤5 members	0.511	[0.99 – 2.80]	0.053
6 – 12 members	-	-	-
Wealth index:			
Poor	-0.143	[0.28 – 2.68]	0.804
Moderate	-0.541	[0.23 – 1.48]	0.257
Rich	-	-	-

Table 4.11 presents the results on the binary logistic regression modelling of having good knowledge of antenatal care services and socio-demographic factors among the newly delivered mothers in selected hospitals in Ondo State, who participated in this study.

For every participant who was living with partner, the odds of having good knowledge of ANC services compared to living with other relatives insignificantly decreases by 24.9% (OR = -0.751, p = 0.227); for every participant who was living with own parents, the odds of having good knowledge of ANC services compared to living with other relatives insignificantly decreases by

25.8% (OR = -0.742, p = 0.312); and for every participant who was living with partner's parents, the odds of having good knowledge of ANC services compared to living with other relatives insignificantly increases by 14.0% (OR = -1.140, p = 0.198).

It can also be seen that for every participant having partner with no formal education, the odds of having good knowledge of ANC services compared to the participant having partner with tertiary education insignificantly increases greatly (OR = 18.2, p = 0.999); for every participant having partner with primary education, the odds of having good knowledge of ANC services compared to the participant having partner with tertiary education insignificantly increases greatly (OR = 18.5, p = 0.999); and for every participant having partner with secondary education, the odds of having good knowledge of ANC services compared to the participant having partner with tertiary education insignificantly decreases by 71.4% (OR = -0.286, p = 0.584).

Results further showed that for every participant having partner who were unemployed/unskilled, the odds of having good knowledge of ANC services compared to the participant having partner who were professionally skilled insignificantly increases by 4.4% (OR = 1.044, p = 0.113); and for every participant having partner who were semi-skilled, the odds of having good knowledge of ANC services compared to the participant having partner who were professionally skilled insignificantly decreases by 34.5% (OR = 0.655, p = 0.227).

Moreover, results showed that for every participant who was in a monogamous marriage, the odds of having good knowledge of ANC services compared to being in a polygamous marriage significantly decreases by 47.5% (OR = -0.525, p = 0.262). Results showed that for every participant whose household members were up to 5, the odds of having good knowledge of ANC services compared to participants whose household size is 6 or more decreases by 48.9% (OR = 0.511, p = 0.053). Results continued to show that for every participant in a poor household, the

odds of having good knowledge of ANC services compared to those in a rich household insignificantly decreases by 85.7% (OR = -0.143, p = 0.804); and for every participant in a moderately rich household, the odds of having good knowledge of ANC services compared to those in a rich household insignificantly decreases by 45.9% (OR = -0.541, p = 0.257).

DISCCUSSION, SUMMARY, CONCLUSION AND RECOMMENDATION

5.1 Discussion of Findings

This research is a study on the determinants of utilization of antenatal care services among newly delivered mothers in selected hospitals in Ondo State, Nigeria. In total, 570 newly delivered mothers who are of reproductive age 15 – 49 who resides in Ondo State were recruited into the study using multi-stage sampling involving, purposive sampling, stratified random sampling and simple random sampling techniques. The scope of the study ran through the three Senatorial Districts of Ondo State and at least one Local Government area was selected to be included in the scope. Overall, six health facilities were used as study hospital sites. The findings also indicated that at least 13% of the recently given birth mothers were enlisted in the Akoko North East Local Government Area, while a smaller proportion (10.5%) were enlisted in the Odigbo Local Government Area. In contrast, the largest proportion (39.1%) of the recently delivered mothers were enlisted in the Akure South Local Government Area. In the same vein, the studies of Titiloye, Womitenren, and Arulogun, (2017), and Atinge, Ogunnowo and Balogun, (2020) also portrayed the use of cross-sectional design to explore the determinants and barriers of antenatal care services utilization among mothers in Ondo and Taraba State, respectively.

Findings showed in this study that the modal age group of newly delivered mothers who participated in this study was 25 – 29 years, while the average age was 31.8 years. The findings indicated that approximately half (50.7%) of the mothers had completed secondary education, while a smaller proportion (38.4%) had reported having tertiary education. However, more than half (52.5%) of the mothers stated that they had semi-skilled occupations, and 25.6% reported having skilled professional jobs. The study also revealed that the vast majority of the participating mothers, around 90%, identified themselves as Christians. Among the Christian participants, 45.8% reported being

Catholics, while 42.3% belonged to other Christian denominations. In contrast, only 11.9% of the mothers identified themselves as Muslims. In comparison, other studies reported a similar modal age group from this study. For example, Sodeinde, Onigbogi, Odukoya, *et al*. (2020) reported modal age group to be 20 – 29 years, with average age of 27.8 years. In Dairo and Atanlogun (2018), the mean age (standard deviation) of the respondents was 21.6+2.1 years. Most of the respondents (80.8%) were within 20-24 years age group, in contrast to this study. Like this study, the majority of the respondents were Yoruba (93.2%), Christians (50.4%) and married (87.1%). About two-fifth of the respondent had secondary education, 85.4% are working predominantly as traders.

The knowledge level of newly delivered mothers about ANC services in selected hospitals in Ondo State, Nigeria

According to the study's main finding, almost 90% of the newly delivered mothers had good knowledge of ANC services and 12% had poor knowledge of ANC services. This finding corroborated with studies in Ilorin, which observed that more than two thirds of the respondents (87.7%) were aware of antenatal care services (Jibril, 2017; Adewoye, Musa, Atoyebi, *et al*., 2019). This study also agreed with the study on the visit of antenatal clinic in Shanghai China (Zhao, Kulane, Gao, *et al*., 2019), which reported that women attending ANC demonstrated a high degree of knowledge about the importance of visiting ANC and breastfeeding of babies.

In contrast, Sodeinke, *et al*. (2020) reported their overall knowledge of the respondent about ANC was low as 46.1% had good knowledge and 53.9% had poor knowledge. In the same vein, Awasthi, Awasthi, Thapa, *et al*. (2018) reported poor knowledge of antenatal care services among their respondents and concluded that poor knowledge of obstetric complication and free delivery services among 38.9% of the respondents was as a result of failure to attend antenatal care clinics. However, in Sumankuuro, Crockett and Wang (2017) suggested in their study that the majority of their

respondents had knowledge of the potential positive value of utilising ANC services, including identification of pregnancy problems and their management through skilled and timely interventions at healthcare facilities. Moreover, this level of knowledge may be attributable to the health education and promotion programmes such as ANC defaulter-tracing, and home visits, when implemented to ensure increased utilisation of maternity services at a health facility. Another contrast was found in Sodeinde, *et al*. (2020) who reported that more than half (53.9%) had poor knowledge about antenatal care services, while 46.9% of the respondents had good knowledge.

The time of initiation of ANC of newly delivered mothers in selected hospitals in Ondo State, Nigeria

According to this study, utilization of antenatal care services was high at 87.5%; and not more than 30% of the mothers registered to receive antenatal care services during their first trimester (<12 weeks of pregnancy), while almost 60% registered during the second trimester or above (≥12 weeks of pregnancy). These findings confirmed the findings of Sodeinde, *et al*. (2020) where 60.0% of the respondents registered for ANC during the second trimester (14-26 weeks) while a few (2.9%) registered in the third trimester. Half (50.0%) of the respondents were interviewed in the second trimester and only 45.5% have had at least four visits at the time of the interview. This study also supported the findings of Fagbamigbe and Idemudia (2017) which found that in all, 6,299 women who had participated in the study, ANC utilization was 65.1%. Moreover, there was a rise in ANC utilization with increased wealth status, from 38.3% among women in the poorest wealth quintile to 92.0% among women in the highest wealth quintile.

In comparison, Geta and Yallew, (2017) found in their study significant variations and inequalities in the proportion of women who initiated ANC contacts during the first trimester across the states in Nigeria. While the proportion who started ANC contacts was high in states such as Benue (44.5%),

Lagos (41.4%) and Nasarawa the levels were lower than 10% in Zamfara, Kano and Sokoto States which achieved less than 2%. Overall, 10 states had 0% – 33% while 27 states attained 34% – 65% ANC initiation during the first trimester. In general, Geta and Yallew, (2017) concluded that the lowest ANC uptake and delayed initiation of ANC contacts among the States in Northern Nigeria compared with the Southern States. Thus, it can be said that the time of initiation of antenatal care services has significantly improved over the years, has confirmed in this study.

Number of ANC visits achieved by newly delivered mothers in selected hospitals in Ondo State, Nigeria

This study found that the majority (87.5%) of the newly delivered mothers had adequate antenatal care visits, 17.9% had first ANC visit during 1 – 3 months of pregnancy. In contribution to these findings, Geta and Yallew, (2017) added on the adequacy of number of ANC visits, only 31.4% of respondents without any formal education visited an ANC provider four or more times; the proportion was 88.1% among their counterparts who had higher education. Current study also supported the findings of Fagbamigbe and Idemudia (2017) which found that ANC utilization was 65.1%, while 56.2% had adequate numbers of ANC visits. Sodeinde, *et al*. (2020) also concluded that the number of antenatal visits and some other maternal variables were significantly associated with clients' satisfaction with antenatal care services. Atinge *et al*. (2020) also reported in their findings that the proportion of women who had 1 or 2-3 ANC visits who delivered in non-facility, 75.0% and 86.7% was significantly higher than the proportion who had 4 or more visits, 69.6%. The number of ANC visits a woman attends was associated with choice of non-facility delivery (p value 0.028).

In contrast to this study, a study in Nepal (Awasthi, Awasthi, Thapa, *et al*., 2018) revealed that 7.0% of them visited ANC once, 3.5% visited twice, 16.7% visited thrice, and the remaining

majority 72.8% percent of them completed all four ANC visits. This finding is consistent with another study done in Nepal where that 70.0% of mothers underwent four antenatal visits, 14.0% did 3 antenatal visits 13.7% with 2 antenatal visits, and 2.1% had a single antenatal visit (Misra & Chaurasia, 2015). Furthermore, among the 114 respondents who attended first ANC visit at a health facility, 51.9% of the respondents had their first antenatal visit at 4 months or later. Similarly, 27.1% of respondents had an antepartum visit in the third month of pregnancy while 21.0% of respondents had their first ANC visit during the second month of pregnancy. Among them who had done their ANC visit, all of the respondent had visited a government hospital. In contrast to this, Awasthi, *et al*. (2018) revealed that 56.0% of the respondents were registered for ANC in the first trimester of pregnancy, whereas only 1.0% were registered in the third trimester. They were registered at different level of health services, i.e., 1.8% in subhealth posts, 28.0% in PHCs, 34.3% in a Zonal hospital, and 35.9% in private nursing homes (Misra & Chaurasia, 2015). This variation must have resulted from the existing good and free maternal and child health services developed for pregnant mother across all public health facilities in the country.

The antenatal care services that are accessible to newly delivered mothers in selected hospitals in Ondo State, Nigeria

Findings in this study showed that the majority (31.2%) of the respondents had received antenatal care services at maternity homes, whereas 25.3% were seen at a government hospital, 25.4% were seen at a government health centre. Furthermore, utilization of antenatal care services was high in this study, while current findings showed that 58.1% of the mothers reported their blood pressure was measured, 75.3% of them reported that their urine sample was taken, 80.5% reported that their blood sample was taken, most (78.1%) of the mothers reported

that they received tetanus toxoid injection, and more than half (57.4%) had reportedly received 2 doses of tetanus toxoid injection. Whereas the majority (80.0%) were given/bought iron and folic acid tablets or iron syrup during last pregnancy.

In comparison to these findings, Atinge *et al*. (2020) expressed that the proportion of women who had ANC in health centre and health post and delivered in a non-facility was 77.6% and 81.0%, respectively. This was more than the proportion of women who had ANC in General Hospital and delivered in a health facility (53.1%). Place of antenatal care was associated with choice of non-facility delivery (p value <0.001). In contrast to this study, Dairo and Atanlogun (2018) reported that of their respondents that attended ANC, more than half (65.4%) of them attended a Primary Health Centre, (5.4%) attended a state specialist hospital, (12.5%) attended a private hospital, (8.5%) attended a teaching hospital (1.2%) attended a maternity home for their antenatal care. PHC has been recommended as the health system strategy to improve access of women to SBAs and ANC, particularly in low-resource settings such as Nigeria (Yaya, Okonofua, Ntoimo, *et al*., 2018).

In contrast to this study, low utilization of antenatal care services components were recorded in a Nigerian study (Agho, Ezeh, Ogbo, *et al*., 2018), where about 63% of mothers received iron/folic supplementation, 60% had had their blood pressure measured, and about 54% had had their blood and urine samples collected for investigations. Less than a quarter of mothers (14%) were given drugs for intestinal parasites, while less than half of mothers had received at least two tetanus toxoid injections, and were also informed about pregnancy complications. Factors strongly associated with underutilization of ANC were rural residence, household economic status, maternal level of education, place of delivery, combined birth order and interval, and lack of access to mass media. The odds of not receiving the complete components of ANC services

were higher in mothers who had had their ANC at home or in a private facility. Other significant factors associated with receipt of inadequate prenatal care content included a desire for pregnancy, distance to health facilities, contraceptive use and insufficient access to mass media. Furthermore, economic status, maternal education and type of residence were also associated with inadequate use of prenatal care content among pregnant women in Nigeria.

The determinants of antenatal care services utilization among newly delivered mothers in selected hospitals in Ondo State, Nigeria

Findings in this study revealed that the uptake of antenatal care services utilization was significantly high. Most importantly, respondents' location distribution were found to be significant determinants of some components of ANC utilization ($p < 0.05$). For example, there was significant association between Senatorial district zoning and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p < 0.0001$). Similarly, there was significant association between Local Government Area and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p < 0.0001$). Furthermore, there was significant association between study hospital site and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p < 0.0001$). Findings also showed that there was significant association between ethnicity and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.009$). There was significant association between size of household and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p < 0.0001$). Similarly, there was significant

association between size of household and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study (p = 0.041). Current findings also revealed that there was significant association between size of household and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.022). Lastly, there was significant association between size of household and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study (p = 0.020).

In contrast to above findings, recent studies have suggested that women in low- and middle-income countries are prone to underutilizing all components of ANC services (Agho, *et al*., 2018; Adewuyi, Auta, Khanal, *et al*., 2018), particularly those considered effective (blood pressure check, blood and urine tests) by WHO (2018). Current study showed that majority of the newly delivered mothers had blood pressure measurements taken and blood and urine tested during their antenatal care during their antenatal visits. This finding is similar to a Ugandan study (Uldbjerg, Schramm, Kaducu, *et al*., 2020) and a study from Southwest Nigeria (Okonofua, *et al*., 2018) which found that only 43% of the respondents had their blood and urine collected for investigations. The low proportion of receipt of adequate antenatal care components observed among pregnant Nigerian women may be attributed to cultural and religious belief, inadequate health facilities and a lack of trained health personnel to conduct recommended tests, particularly in the rural communities.

Findings suggested that receiving ANC services from a home facility was associated with inadequate receipt of ANC components compared with receiving ANC from a government facility. This finding was not surprising as only one-third of women of childbearing age have facility births in Nigeria (Onoh, Egede, Lawani, *et al*., 2020). A possible reason for the high

preference of home-based birth or care may be attributed to cultural beliefs, affordability,36 convenience, previous uncomplicated pregnancies (Xu, Peng, Ahuru, *et al*., 2020) and fear of medical procedures, such as blood collection for investigations and pelvic examinations during delivery (Shee, Frawley, Robertson, *et al*., 2021). Overcoming these barriers will take considerable effort at the individual, household and community levels, as well as commitment from all levels of government in Nigeria. Additionally, integrating the WHO's deworming programme into ANC, shown to be successful elsewhere (Salam, Das & Bhutta, 2021), as an adjunct to those interventions will also improve the quality of ANC in Nigeria.

5.2 Summary of Findings

The study was conducted in six selected health facilities in Ondo State, Nigeria with the sole aim to identify the determinants of utilization of antenatal care services among newly delivered mothers. A total of 570 newly delivered mothers, aged between 15 and 49 years, residing in Ondo State, were selected to participate in the study. The selection process involved various sampling techniques, including purposive sampling, stratified random sampling, and simple random sampling. The study covered all three Senatorial Districts of Ondo State, with at least one Local Government area included in the scope. Six health facilities were chosen as study hospital sites.

This thesis explored the research questions and objectives stated above in relation to all women who had recently been delivered in the selected hospital facilities in Ondo State during the time period of the study. Furthermore, the data collection and analysis period of this thesis were delimited to two to three months after the proposal was approved by the study supervisor. Although this thesis investigated the utilization of antenatal care services and their determinants in hospitals in Ondo

State, the respondents were sampled and surveyed among recently delivered mothers who may or may not have utilized any antenatal care services during their pregnancy. This was intended to increase the scope and reduce the delimitation of the study within a reasonable period of time. A descriptive cross-sectional community-based methodology was used in this research. The target population of this study consists of newly delivered mothers who are of reproductive age 15 – 49 who resides in Ondo State. Most importantly, the instrument was adapted from the Nigerian Demographic and Health Survey (NDHS) questionnaire, using the sections on Basic Characteristics of Survey Respondents, Antenatal Care Coverage and Content, Components of ANC Visits, and Delivery Services.

The findings revealed that at least 13% of the recently given birth mothers were recruited from the Akoko North East Local Government Area, while a smaller proportion (10.5%) were recruited from the Odigbo Local Government Area. In contrast, the largest proportion (39.1%) of the recently delivered mothers were recruited from the Akure South Local Government Area. Results showed that the mean age of newly delivered mothers who participated in this study was 31.8 years with standard deviation of 8.9 years. Meanwhile, modal age group (23.0%) was found to be at 25 – 29 years. However, about half (50.7%) of the mothers had secondary education, while even less (38.4%) reportedly had tertiary education. Furthermore, results showed that the majority (74.7%) of the mothers were of the Yoruba tribe, 17.5% were of the Igbo tribe and 6.1% were of the Hausa tribe. It was further revealed that 8.6% of these newly delivered mothers reported that the location of last delivery was at home or other homes (1.8%). However, 27.4% of the mothers reported that they had delivered at a Government hospital, majority (47.7%) had reportedly delivered at a Government health centre.

Results showed that about 88% of the newly delivered mothers had good knowledge of ANC services and 12% had poor knowledge of ANC services in selected hospitals in Ondo State, Nigeria. Findings in this study revealed that the uptake of antenatal care services utilization was significantly high (87.5%). Most importantly, respondents' location distribution were found to be significant determinants of some components of ANC utilization ($p < 0.05$). For example, there was significant association between Senatorial district zoning and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p < 0.0001$). Similarly, there was significant association between Local Government Area and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p < 0.0001$). Furthermore, there was significant association between study hospital site and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p < 0.0001$). Findings also showed that there was significant association between ethnicity and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.009$). There was significant association between size of household and pattern of blood pressure measurement during antenatal care visits among newly delivered mothers who participated in this study ($p < 0.0001$). Similarly, there was significant association between size of household and pattern of blood sample taken during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.041$). Current findings also revealed that there was significant association between size of household and pattern of tetanus injection received during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.022$). Lastly, there was significant association between size of household and pattern of Iron & Folic acid tabs/syrup received during antenatal care visits among newly delivered mothers who participated in this study ($p = 0.020$).

5.3 Conclusion

This study indicated that location, low household wealth and longer distance to a health facility were strong predictors of underutilization of ANC services in Nigeria. These findings indicate the need to implement both community and individual level interventions. At the community level, the establishment of a strong link between the community and health facilities by using change agents, such as community health workers, community leaders and religious, traditional and social group leaders, will increase the utilization of ANC services in Nigeria. At the individual level, financial incentives (such as government subsidies for ANC services or free ANC services) and advocacy for ANC importance among women, especially those from low socioeconomic status groups and rural areas, will maximize outcomes. Improvement in the primary healthcare delivery system through the National Primary Health Care Development Agency will also improve the uptake and quality of ANC in Nigeria.

Nigeria, like other developing countries, is lagging behind in ANC utilization and disproportionately bears the burden of maternal death and ill health compared with the developed countries. Ironically, ANC use is lowest among women who need it most. These are the poor, uneducated women living in rural areas. Despite the prevalent free maternal health care policy in Nigeria, we found poverty and poor educational attainment to have influenced ANC utilization and adequacy of number of ANC visits more than any other factor. These have created health inequalities, a societal problem that must be addressed through effective systems because it undermines the economy, heightens social costs, and reduces overall well-being. Although some state governments in Nigeria had introduced and implemented free ANC policy, it is alien in some states. There are other direct and indirect opportunity costs, however, associated with ANC use with which the expectant mothers have to contend. Beyond free maternal and child health

care, poverty must be dealt with at individual and household levels in Nigeria for the country to compete favourably with other countries as far as ANC utilization is concerned.

5.4 Limitations

The current study has some limitations. Recall bias might have set in as some respondents were illiterate and most respondents had to recall when they initiated ANC contacts, except for few cases that were found on ANC cards. The identified factors were only associated with the timing of ANC contacts and should not be interpreted as the causes. Our choice of explanatory variables was limited by the secondary nature of the data. We assumed that there are primary healthcare facilities in each locality or area in Nigeria where ANC services are accessible. However, this may not be so. The closeness and availability of healthcare facilities and women's access to them can influence the initiation of ANC contact.

5.5 Recommendations

It is crucial for local, state, and national governments to adopt and implement the WHO recommendations regarding the utilization of antenatal care (ANC), particularly focusing on initiating ANC contacts during the first trimester of pregnancy. Initiating ANC early serves as a solution for achieving the optimal number of ANC visits. To address the practice of late ANC initiation in Nigeria, collaborative efforts from all stakeholders in maternal and child healthcare are necessary. Encouraging early enrolment in ANC services and raising awareness among the general population for optimal ANC utilization should be prioritized. Interventions aimed at improving ANC attendance should also address barriers associated with inequities and inequalities in ANC utilization, particularly considering the geographical location of residence. Currently, there is a disparity between the North and the South, with the North lagging behind in

early ANC initiation. ANC stakeholders in the Northern states should learn from the successful practices implemented in the Southern states where early initiation rates are higher. Although this study did not assess the existing health systems in Nigeria, it is essential for the government and other stakeholders at the state level to ensure that the health system is adequately equipped to meet the increased demands that may arise from early ANC initiation in Nigeria. Additionally, further research is needed to examine the capacity of the Nigerian health system and its influence on the timing of ANC initiation.